Faculty of Science
Department of Mathematics
VU

Linear Algebra
Final Exam 2023
29-03-2023, 8:30–11:15

Name: ______________________________

Student ID: ______________________________

Question:	1	2	3	4	5	6	7	Total
Points:	7	7	5	7	5	3	2	36
Score:								

Grade: ______

$$\text{Grade} = \frac{\#\text{total points}}{4} + 1$$

Read the instructions carefully!

- The use of a calculator, book or notes is **not** allowed.
- Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page, but make sure the final answer is on the front.
- Answers without supporting work will not be given credit.

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1. Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ and $T : \mathbb{R}^5 \to \mathbb{R}^4$ be a linear transformation given by

$$T(\mathbf{x}) = \begin{bmatrix} x_1 + 2x_3 + x_4 \\ -x_1 + x_2 + x_3 + x_5 \\ -x_2 - x_3 + 2x_4 + x_5 \\ x_1 + x_2 - x_4 - x_5 \end{bmatrix}.$$

(a) **(1 pts.)** What is the standard matrix A of T?

(b) **(3 pts.)** Find a basis for the Column Space of A.

(c) **(3 pts.)** Find a basis for the Null Space of A.

2. Let $B = \begin{bmatrix} -1 & 2 & -1 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix}$.

(a) **(2 pts.)** Determine the LU-factorization of B.

(b) **(3 pts.)** Find the inverse of B.

(c) **(2 pts.)** Let $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$. Find $[\mathbf{x}]_{\mathcal{B}}$, where $\mathcal{B}$ is the basis made out of the columns of B.

3. Let

$$H = \left\{ \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \\ \mathbf{x_3} \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x_1} + \mathbf{x_2} + \mathbf{x_3} = 0 \right\}.$$

(a) **(3 pts.)** Prove that H is a subspace of $\mathbb{R}^3$.

(b) **(2 pts.)** Find a basis of H.

4. Let

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}.$$

(a) **(1 pts.)** Show that $\mathbf{v_1} = \begin{bmatrix} 3 \\ -11 \\ 10 \end{bmatrix}$ is an eigenvector of A and find its corresponding eigenvalue λ_1.

(b) **(4 pts.)** Show that the remaining eigenvalues of A are $\lambda_2 = 2$ and $\lambda_3 = 6$, and give a basis for every eigenspace of A.

(c) **(2 pts.)** Determine whether A is diagonalizable. If so, give the diagonalization, i.e., an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. If not, explain why A is not diagonalizable.

5. Let W be a subspace of $\mathbb{R}^4$ spanned by $\mathbf{w_1} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{w_2} = \begin{bmatrix} -1 \\ 1 \\ -3 \\ 3 \end{bmatrix}$, $\mathbf{w_3} = \begin{bmatrix} 2 \\ 0 \\ -4 \\ 2 \end{bmatrix}$.

(a) **(3 pts.)** Find an orthogonal basis for W.

(b) **(2 pts.)** Let $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \notin W$. Find the best approximation of $\mathbf{y}$ by vectors of W.

6. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

   (a) **(1 pts.)** The map T defined by $T((x_1, x_2)) = (x_1 + x_2, x_1 - x_2, x_2)$ is linear.

   (b) **(1 pts.)** If A is a 3×4 matrix, then the equation $Ax = 0$ has only the trivial solution.

   (c) **(1 pts.)** Let A be a square matrix. If all the eigenvalues of A are zero, then A is the zero matrix.

7. **(2 pts.)** Let

$$A_2 = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix},$$

$$A_3 = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 3 \\ -1 & -2 & 0 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -1 & 0 & 3 & 4 \\ -1 & -2 & 0 & 4 \\ -1 & -2 & -3 & 0 \end{bmatrix},$$

$$A_n = \begin{bmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ -1 & 0 & 3 & \cdots & n-1 & n \\ -1 & -2 & 0 & \cdots & n-1 & n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -2 & -3 & \cdots & 0 & n \\ -1 & -2 & -3 & \cdots & -n+1 & 0 \end{bmatrix}$$

with $n > 1$. Show that $\det A_n = n!$, where $n! = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot n-1 \cdot n$.
Note: Do not use induction on n, but rather calculate the determinant of A_n.
Hint: Make use of the properties of the determinant of a matrix on row operations.